

We can use Ward identity to show that $\delta_3 = 0$ to all orders in perturbation theory.

Let's start with one loop contribution

at 1-loop we have

$$\text{Diagram 1} = \text{Diagram 2} + \text{Diagram 3}$$

Diagram 1: A fermion line with a loop (shaded circle) and an external photon line with momentum q.

Diagram 2: A fermion line with a loop (circle with a dot) and an external photon line with momentum q.

Diagram 3: A fermion line with a loop (circle with a cross) and an external photon line with momentum q.

with $\text{Diagram 1} = i\Pi_2^{\mu\nu}(q) = (q^2 g^{\mu\nu} - q^\mu q^\nu) \Pi_2(q^2)$

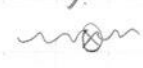
and $\text{Diagram 3} = -i(q^\mu g^{\mu\nu} - q^\mu q^\nu) \delta_3 - \delta_3 q^\mu q^\nu$

Ward-Takahashi symmetry tells us that

$$q_\mu \text{Diagram 1} = 0$$

$$\Rightarrow q_\mu (-\delta_3 q^\mu q^\nu) = 0 \quad \text{must be true since } q_\mu (q^\mu g^{\mu\nu} - q^\mu q^\nu) = 0$$

$$\Rightarrow \delta_3 = 0 \quad \text{to the first order in perturbation theory.}$$

The gauge fixing term only appears in the 1-loop counter-term and because of ward identity we saw that $\delta_3 = 0$ in first order. The Ward identity also forces δ_3 to be zero in all orders in perturbation theory because higher order terms get δ_3 only from the diagram  for which Ward identity gives $\delta_3 = 0$.

Exercise 3.9

Prove Furry's theorem for QED.

Furry's theorem: Vacuum expectation value of any odd number of electromagnetic currents must vanish.

The vacuum expectation value ^{N-currents} is proportional to

$$\langle \Omega | T(j_\mu(x_1) \dots j_\mu(x_n)) | \Omega \rangle \equiv V$$

So if $V = 0$ then the ^{vacuum} expectation value is 0

Charge conjugation C (with $C^2 = 1$) is a symmetry of QED which means $C|\Omega\rangle = |\Omega\rangle$ (and also $\langle\Omega|C^\dagger = \langle\Omega|$).

Let's now insert $1 = C^\dagger C$ around j_μ 's in V

with the denominator

$$D = x[(p-k)^2 + i\epsilon] + y[(k+q)^2 - m^2 + i\epsilon] + z[k^2 - m^2 + i\epsilon]$$

$$\begin{cases} x+y+z=1 \\ p^2=m^2 \\ p'^2=m^2 \end{cases}$$

$$= x p^2 - 2xp \cdot k + \underline{x k^2} + \underline{y k^2} + 2yk \cdot q + yq^2 - ym^2 + \underline{z k^2} - zm^2 + i\epsilon(x+y+z)$$

$$= k^2 + 2(yq - xp) \cdot k + yq^2 - m^2(-x+y+z) + i\epsilon \quad \parallel q = p' - p$$

$$= k^2 + 2(y p' - (x+y)p) \cdot k + yq^2 - m^2(-x+y+z) + i\epsilon$$

Substitute:

$$= (k + (y p' - (x+y)p))^2 - (y p' - (x+y)p)^2 + yq^2 - m^2(-x+y+z) + i\epsilon$$

$$\begin{cases} l = k + (y p' - (x+y)p) \\ \cancel{p} l = k + (y p' - (1-z)p) \end{cases}$$

$$= l^2 - (y^2 p'^2 + (1-z)^2 p^2 + 2 \cdot y(1-z)p' \cdot p) + yq^2 - m^2(-x+y+z) + i\epsilon$$

$$\begin{cases} \text{Since } x+y=1-z \\ \parallel p^2=m^2=p'^2 \end{cases}$$

$$= l^2 - y(1-z) \underbrace{[p^2 + p'^2 - (p' - p)^2]}_{= 2 p' \cdot p} + yq^2 - m^2[y^2 + (1-z)^2 - x+y+z] + i\epsilon \quad \parallel q = p' - p$$

$$= l^2 - y(1-z)[2m^2 - q^2] + yq^2 - m^2[y^2 + (1-z)^2 - x+y+z] + i\epsilon$$

$$= l^2 + yq^2[1+1-z] - m^2[2y(1-z) + y^2 + (1-z)^2 - x+y+z] + i\epsilon$$

$$= l^2 + y(x+y)q^2 - m^2[(1-z)(1-z+2y) + y(1+y) - x+z]$$

$$= l^2 - \Delta + i\epsilon \quad \text{with } \Delta \equiv -y(x+y)q^2 + m^2[(1-z)(1-z+2y) + y(1+y) - x+z]$$

Now let's simplify the Numerator N^μ to find $F_2(q^2, m)$

$$N^\mu \equiv \gamma_\nu (\not{k} + m) \gamma^\mu (\not{k}' + m) \gamma^\nu, \quad \text{we will be using dimensional regularization}$$

So the Dirac algebra will be in D -dimensions: $\gamma^\mu \gamma_\mu = D$ and $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$

$$\Rightarrow \gamma^\nu \not{k} \gamma_\nu = p_\mu \gamma^\nu \gamma^\mu \gamma_\nu = p_\mu (2g^{\mu\nu} - \gamma^\mu \gamma^\nu) \gamma_\nu = 2\not{k} - D\not{k}$$

$$\Rightarrow \gamma^\nu \not{k} \gamma_\nu = \underline{-2\not{k} + (4-D)\not{k}} \quad \begin{cases} \rightarrow \gamma^\nu \not{k} \gamma^\mu \not{k} \gamma_\nu = -2\not{k}' \gamma^\mu \not{k} + (4-D)\not{k} \gamma^\mu \not{k}' \\ \rightarrow \gamma^\nu \not{k} \gamma^\mu \gamma_\nu = 4\not{k}^\mu - (4-D)\not{k} \gamma^\mu \end{cases}$$

For other contraction we can use equation from Peskin (7.89), by replacing $\epsilon = 4-D$

We can now simplify N^μ

$$N^\mu = \gamma_\nu (\not{k} + m) \gamma^\mu (\not{k}' + m) \gamma^\nu = \gamma^\nu \not{k} \gamma^\mu \not{k}' \gamma_\nu + m \gamma^\nu \not{k} \gamma^\mu \gamma_\nu + m \gamma^\nu \gamma^\mu \not{k}' \gamma_\nu + m^2 \gamma^\nu \gamma^\mu \gamma_\nu$$

Before going forward let's remind ourselves that we are only looking for $F_2(q^2, m^2)$

Which means we can drop terms that are proportional to a single Dirac matrix $\gamma^\mu \Rightarrow$ so we can drop $m \gamma^\nu \gamma^\mu \gamma_\nu$ but terms like $\not{p} \gamma^\mu$ or $\gamma^\mu \not{p}$ we must evaluate properly. Using the contractions we get:

$$\Rightarrow N^\mu = -2 \not{k}' \gamma^\mu \not{k} + (4-D) \not{k} \gamma^\mu \not{k}' + 4m [k + k']^\mu - m(4-D) (\not{k} \gamma^\mu + \gamma^\mu \not{k}')$$

Let's now substitute $k' = k + q$

$$\Rightarrow N^\mu = -2(\not{k} + \not{q}) \gamma^\mu \not{k} + (4-D) \not{k} \gamma^\mu (\not{k} + \not{q}) + 4m(2k + q)^\mu - m(4-D) (\not{k} \gamma^\mu + \gamma^\mu \not{k} + \gamma^\mu \not{q})$$

$$N^\mu = -2(\not{k} + \not{q}) \gamma^\mu \not{k} + (4-D) \not{k} \gamma^\mu (\not{k} + \not{q}) + 4m(2k + q)^\mu - m(4-D) [\not{k} \gamma^\mu + \gamma^\mu \not{k} + \gamma^\mu \not{q}]$$

Let's now do the shift $\underline{l} = k + y p' - (1-z)p$

Trick: We can use $\bar{u}(p') \not{p}' = \bar{u}(p') m$ and $\not{p} u(p) = m u(p)$ because Γ^μ comes in vertex in the form of $i e \bar{u}(p') \Gamma^\mu u(p)$

$$\Rightarrow N^\mu = -2(\not{k} - y \not{p}' + (1-z)\not{p} + \not{q}) \gamma^\mu (\not{k} - y \not{p}' + (1-z)\not{p}) + (4-D) (\not{k} - y \not{p}' + (1-z)\not{p}) \gamma^\mu (\not{k} - y \not{p}' + (1-z)\not{p} + \not{q}) - m(4-D) [(\not{k} - y \not{p}' + (1-z)\not{p}) \gamma^\mu + \gamma^\mu (\not{k} - y \not{p}' + (1-z)\not{p}) + \gamma^\mu \not{q}] + 4m (2(\not{k} - y \not{p}' + (1-z)\not{p}) + \not{q})^\mu$$

All of terms with $\underline{l}$ vanish Reasoning below:

$$\Rightarrow N^\mu = -2(-y \not{p}' + (1-z)\not{p} + \not{q}) \gamma^\mu (-y \not{p}' + (1-z)\not{p}) + (4-D) (-y \not{p}' + (1-z)\not{p}) \gamma^\mu (-y \not{p}' + (1-z)\not{p} + \not{q}) - m(4-D) [(-y \not{p}' + (1-z)\not{p}) \gamma^\mu + \gamma^\mu (-y \not{p}' + (1-z)\not{p}) + \gamma^\mu \not{q}] + 4m (2(-y \not{p}' + (1-z)\not{p}) + \not{q})^\mu$$

The term with odd number of power of $\underline{l}$ in the numerator vanishes by symmetric integration $\Rightarrow$ we can drop the terms linear in $\underline{l}$ from N^μ

furthermore in the D -dimensional integral $\int d^D l$ we can write

$\not{k} \gamma^\mu \not{k} = \gamma^\nu \gamma^\mu \gamma^\rho l_\nu l_\rho = \gamma^\nu \gamma^\mu \gamma^\rho g_{\nu\rho} \frac{1}{D} l^2 \sim \gamma^\mu$ so we can drop it since we are looking to find $F_2(q, m^2)$. With this we can drop all $\underline{l}$ terms.

Let's now use the trick: $\bar{u}(p')\not{p}' = \bar{u}(p')m$ and $\not{p}u(p) = m u(p)$ again

$$N^{\mu} = -2[-ym + (1-z)\not{p} + \not{q}]\gamma^{\mu}[-y\not{p}' + (1-z)m] \\ + (4-D)[-ym + (1-z)\not{p}]\gamma^{\mu}[-y\not{p}' + (1-z)m + \not{q}] \\ - m(4-D)[\underbrace{(-ym + (1-z)\not{p})}_{\text{can drop}}\gamma^{\mu} + \gamma^{\mu}(-y\not{p}' + \underbrace{(1-z)m}_{\text{can drop}}) + \gamma^{\mu}\not{q}] \\ + 4m(-2yp' + 2(1-z)p + q)^{\mu}$$

Now let's use $q = p' - p \Leftrightarrow p = p' - q \Leftrightarrow p' = p + q$

$$\Rightarrow N^{\mu} = -2[-ym + (1-z)\overset{\text{from } p'}{m} - (1-z)\not{q} + \not{q}]\gamma^{\mu}[-y\overset{\text{from } p'}{[q+m]} + (1-z)m] \quad || \quad 1-y-z=x \\ + (4-D)[-ym + (1-z)m - (1-z)\not{q}]\gamma^{\mu}[-y[q+m] + (1-z)m + \not{q}] \\ - m(4-D)[\underbrace{(1-z)(m-q)}_{\text{drop}}\gamma^{\mu} + \gamma^{\mu}(-y(\underbrace{m+q}_{\text{drop}}) + \not{q})] + 4m(-2yp' + 2(1-z)p + q)^{\mu}$$

$$N^{\mu} = -2(xm + zq)\gamma^{\mu}(xm - yq) + 4m(-2yp' + 2(1-z)p + q)^{\mu} \\ + (4-D)[xm - (1-z)q]\gamma^{\mu}[xm + (1-y)q] \\ - m(4-D)[-(1-z)q\gamma^{\mu} + (1-y)\gamma^{\mu}q]$$

Now let's simplify some terms:

$$(xm + zq)\gamma^{\mu}(xm - yq) = \underbrace{x^2m\gamma^{\mu}}_{\text{drop}} + xmzq\gamma^{\mu} - xmy\gamma^{\mu}q - \underbrace{zyq\gamma^{\mu}q}_{\text{drop}} \quad || \quad 1 = x+y+z$$

$$= xmzq\gamma^{\mu} - xmy\gamma^{\mu}q = xm(1+z)q\gamma^{\mu} - xm\gamma^{\mu}qx + xm\gamma^{\mu}q - xm(1-y)\gamma^{\mu}q$$

$$= xm[\gamma^{\mu}, q] + x\frac{m}{2}(2+z)q\gamma^{\mu} + x\frac{m}{2}(2y-2)\gamma^{\mu}q \quad || \quad 1 = x+y+z$$

$$= xm[\gamma^{\mu}, q] + \frac{m}{2}x(2+z+z-y+y)q\gamma^{\mu} + x\frac{m}{2}(y+y-2-z+z)\gamma^{\mu}q$$

$$= xm[\gamma^{\mu}, q] + \frac{m}{2}x(z-y)[q\gamma^{\mu} - \gamma^{\mu}q] + \frac{m}{2}x[2+z+y]q\gamma^{\mu} + \frac{m}{2}x[-2-z+y]\gamma^{\mu}q$$

$$= \cancel{xm[\gamma^{\mu}, q]} + \frac{m}{2}x(z-y)[q, \gamma^{\mu}] + \cancel{xm[q, \gamma^{\mu}]} + \frac{m}{2}x(z+y)\{q, \gamma^{\mu}\} \quad || \quad \begin{cases} \{q, \gamma^{\mu}\} = \{\gamma^{\mu}, q\} \\ [q, \gamma^{\mu}] = -[\gamma^{\mu}, q] \end{cases}$$

$$= \frac{m}{2}x(z-y)[q, \gamma^{\mu}] + \frac{m}{2}x(z+y)\{q, \gamma^{\mu}\}$$

We now use identity $\frac{1}{2}[\gamma^{\mu}, q] = -i\sigma^{\mu\nu}q_{\nu}$ and $\frac{1}{2}\{q, \gamma^{\mu}\} = q^{\mu}$

$$\Rightarrow (xm+zq)\gamma^{\mu}(xm-yq) = \underline{imx(z-y)\sigma^{\mu\nu}q_{\nu} + mx(z+y)q^{\mu}}$$

And now we look at $(xm - (1-z)q)\gamma^{\mu}(xm + (1-y)q) = \underbrace{x^2m\gamma^{\mu}}_{\text{drop}} - xm(1-z)q\gamma^{\mu} + xm(1-y)\gamma^{\mu}q - \underbrace{q\gamma^{\mu}q}_{\text{drop}}$

$$\Rightarrow (xm - (1-z)q)\gamma^{\mu}(xm + (1-y)q) = -xm(1-z)q\gamma^{\mu} + xm(1-y)\gamma^{\mu}q = -x\frac{m}{2}(2-z-z+y-y)q\gamma^{\mu} + \frac{xm}{2}(2-2y-z+z)\gamma^{\mu}q$$

$$= xm\frac{1}{2}(z-y)(q\gamma^{\mu} + \gamma^{\mu}q) + x\frac{m}{2}(z+y-2)(q\gamma^{\mu} - \gamma^{\mu}q)$$

$$= \frac{m}{2}x(z-y)\{q, \gamma^{\mu}\} + \frac{m}{2}x(\underbrace{2-z-y}_{=1+x})[\gamma^{\mu}, q] \quad || \quad \frac{1}{2}[\gamma^{\mu}, q] = -i\sigma^{\mu\nu}q_{\nu}, \quad \frac{1}{2}\{q, \gamma^{\mu}\} = q^{\mu}$$

$$= \underline{X(z-y)m q^{\mu} - i\sigma^{\mu\nu}q_{\nu} X(1+x)m}$$

and finally $-(1-z)q\gamma^\mu + (1-y)\gamma^\mu q$

$$\begin{aligned} -(1-z)q\gamma^\mu + (1-y)\gamma^\mu q &= -\frac{1}{2}(2-z-z+y-y)q\gamma^\mu + \frac{1}{2}(2-y-y+z-z)\gamma^\mu q \\ &= \frac{1}{2}\underbrace{(2-z-y)}_{=1+x}[-q\gamma^\mu + \gamma^\mu q] + \frac{1}{2}(z-y)[q\gamma^\mu + \gamma^\mu q] \\ &= (1+x)\frac{1}{2}[\gamma^\mu, q] + (z-y)\frac{1}{2}\{\gamma^\mu, q\} \quad \parallel \begin{aligned} \frac{1}{2}[\gamma^\mu, q] &= -i\sigma^{\mu\nu}q_\nu \\ \frac{1}{2}\{\gamma^\mu, q\} &= q^\mu \end{aligned} \\ &= -i(1+x)\sigma^{\mu\nu}q_\nu + (z-y)q^\mu \end{aligned}$$

Now our N^μ in terms of these is :

$$\begin{aligned} N^\mu &= -2[imx(z-y)\sigma^{\mu\nu}q_\nu + mx(z+y)q^\mu] + 4m(-2yp' + 2(1-z)p + q)^\mu \\ &\quad + (4-D)[x(z-y)mq^\mu - i\sigma^{\mu\nu}q_\nu x(1+x)m] \\ &\quad - m(4-D)[-i(1+x)\sigma^{\mu\nu}q_\nu + (z-y)q^\mu] \end{aligned}$$

$$\begin{aligned} N^\mu &= m(4-D)[+i\sigma^{\mu\nu}q_\nu(-x(1+x) + (1+x)) + q^\mu(x(z-y) - (z-y))] \\ &\quad + 4m[-2yp' + 2(1-z)p + p' - p]^\mu \\ &\quad - 2m[ix(z-y)\sigma^{\mu\nu}q_\nu + x(z+y)q^\mu] \end{aligned}$$

$$\begin{aligned} N^\mu &= mi\sigma^{\mu\nu}q_\nu[(4-D)(1+x)(1-x) - 2x(z-y)] \\ &\quad + m q^\mu[(4-D)(x-1)(z-y)] + 4m[-(1-2y)p' + (1-2z)p]^\mu \end{aligned}$$

$$\begin{aligned} (1-2y)p' + (1-2z)p &= (1-y-y+z-z)p' + (1-z-z+y-y)p \\ &= (z-y)(p' - p) + (1-z-y)(p' + p) \\ &= (z-y)q + x(p' + p) \end{aligned}$$

$$\Rightarrow [(1-2y)p' + (1-2z)p]^\mu = (z-y)q^\mu + x(p' + p)^\mu$$

using Gordon decomposition we get $(p' + p)^\mu = \underbrace{2m\gamma^\mu}_{\text{we drop this as before}} - i\sigma^{\mu\nu}q_\nu$

$$\Rightarrow N^\mu = i\sigma^{\mu\nu}q_\nu m \left[\frac{1}{2}(4-D)(1+x)(1-x) - x(z-y) \right] + m q^\mu [(4-D)(x-1)(z-y)] + 4m(z-y)q^\mu + 4mx(-i\sigma^{\mu\nu}q_\nu)$$

$$N^\mu = i\sigma^{\mu\nu}q_\nu m [-4x - x(z-y) + (4-D)(1+x)(1-x)] + m q^\mu [(4-D)(x-1) + 4](z-y)$$

the term $m q^\mu (z-y) [(4-D)(x-1) + 4]$ is odd under $x \leftrightarrow y$

So it integrates to zero, same as in lecture slide (week 3 part 2) page 8.

And all that remains is now the term proportional to $i\sigma^{\mu\nu}q_\nu$

$$\Rightarrow N^\mu = \frac{i\sigma^{\mu\nu}q_\nu}{2m} \cdot 2m^2 \left[-x(4+z-y) + (4-D)(1-x^2) \right]$$

Since $ie\Gamma^\mu(p', p) = ie(F_1(q^2)\gamma^\mu + \frac{i\sigma^{\mu\nu}q_\nu}{2m}F_2(q^2))$

and the $ie\Gamma^\mu$ with only the $\frac{i\sigma^{\mu\nu}q_\nu}{2m}$ part is

$$(-ie)^3 i \int_0^1 dx dy dz \delta(x+y+z-1) \int \frac{d^D l}{(2\pi)^D} \frac{2N^\mu}{\tilde{D}^3}, \text{ from the beginning of exercise}$$

with $\tilde{D} = (l^2 - \Delta + i\epsilon)$ with $\Delta = -y(x+y)q^2 + m^2[(1-z)(1-z+2y) + y(1+y) \overset{\uparrow}{x+z}]$

$$\Rightarrow \frac{i\sigma^{\mu\nu}q_\nu}{2m} F_2(q^2) = -(-ie)^2 i \int_0^1 dx dy dz \delta(x+y+z-1) \int \frac{d^D l}{(2\pi)^D} \frac{2N^\mu}{\tilde{D}^3}$$

$$\stackrel{\downarrow}{=} 2ie^2 \frac{i\sigma^{\mu\nu}q_\nu}{2m} \int_{x,y,z} \delta(x+y+z-1) \int \frac{d^D l}{(2\pi)^D} \frac{2m^2 [-x(4+z-y) + (4-D)(1-x^2)]}{(l^2 - \Delta + i\epsilon)^3}$$

$$\Rightarrow F_2(q^2) = 4ie^2 m^2 \int_{x,y,z} \delta(x+y+z-1) [-x(4+z-y) + (4-D)(1-x^2)] \int \frac{d^D l}{(2\pi)^D} \frac{1}{(l^2 - \Delta + i\epsilon)^3}$$

This is the expression for $F_2(q^2, m^2)$.

Let's now calculate $F_2(0, m^2)$

Let's first calculate the D -dimensional integral using

Peskin (A.44)

$$\Rightarrow \int \frac{d^D l}{(2\pi)^D} \frac{1}{(l^2 - \Delta)^3} = \frac{(-1)^3 i}{(4\pi)^{D/2}} \frac{\Gamma(3 - \frac{D}{2})}{\Gamma(3)} \left(\frac{1}{\Delta}\right)^{3 - \frac{D}{2}}$$

$$\Rightarrow F_2(0, m^2) = 4ie^2 m^2 \int_{x,y,z} \delta(x+y+z-1) [-x(4+z-y) + (4-D)(1-x^2)] \frac{-i}{(4\pi)^{D/2}} \frac{\Gamma(3 - \frac{D}{2})}{\Gamma(3)} \left(\frac{1}{\Delta(q^2=0)}\right)^{3 - \frac{D}{2}}$$

We notice that there is no divergence for $D=4$ so we can just set $D=4$

$$\Rightarrow F_2(0, m^2) = \frac{4e^2 m^2}{(4\pi)^2} \int_{x,y,z} \delta(x+y+z-1) [-x(4+z-y)] \frac{\Gamma(1)}{\Gamma(3)} \left(\frac{1}{\Delta(q^2=0)}\right) \parallel \begin{matrix} \Gamma(1)=1 \\ \Gamma(3)=2 \end{matrix}$$

$$= \frac{e^2 m^2}{8\pi^2} \int_{x,y,z} \delta(x+y+z-1) \frac{-x(4+z-y)}{m^2 [(1-z)(1-z+2y) + y(1+y) - x+z]} \parallel \begin{matrix} \alpha = \frac{e^2}{4\pi} \\ 4 = 3 + x + z + y \end{matrix}$$

$$= \frac{\alpha}{2\pi} \int_0^1 dx \int_0^{1-x} dz \frac{-x(3+2z+x)}{[(1-z)(3-3z-2x) + (1-x-z)(2-x-z) - x+z]} \parallel \begin{matrix} \text{Let's calculate the} \\ \text{Integral with calculator} \\ \text{which gives } 1 \end{matrix}$$

$$\Rightarrow \underline{\underline{F_2(0, m^2) = \frac{\alpha}{2\pi}}} \quad \square$$

This could be simplified further but will leave it as it is.

Would still be bit different than in lecture note example since the momentum label in the notes are different to ones in tutorial which I am using. Also tutorial/hints quotes the lecture note result so actual result with tutorials momentum labels is different. This is also why my N^μ looks different to one in lecture note.